



PROBABILITY THEORY

MTM2592-Gr:2

FIFTH WEEK

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MIXED RANDOM VARIABLES

Definition: Dirac delta function is defined as

$$\delta(x - x_0) = \begin{cases} \infty, & x = x_0 \\ 0, & x \neq x_0 \end{cases}$$

with the condition

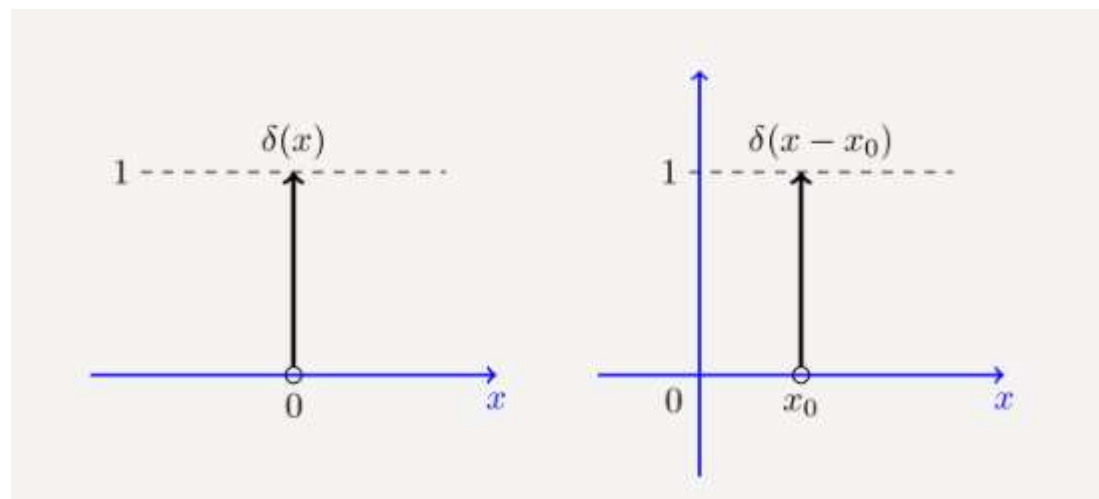
$$\int_{x_0 - \varepsilon_0}^{x_0 + \varepsilon_0} \delta(x - x_0) dx = 1 \quad \text{for any } \varepsilon > 0.$$

$\delta(x - x_0)$ is not a traditional function and no function defined on the real numbers has the properties given above. It can be considered as a probability distribution. It also satisfies;

$$1) \quad \delta(x - x_0) = \frac{d}{dx} u(x - x_0) \quad \text{where } u(x - x_0) = \begin{cases} 1, & \text{if } x \geq x_0 \\ 0, & \text{if } x < x_0 \end{cases}$$

$$2) \quad \int_{x_0 - \varepsilon_0}^{x_0 + \varepsilon_0} g(x) \delta(x - x_0) dx = \int_{-\infty}^{\infty} g(x) \delta(x - x_0) dx = g(x_0)$$

For any $\epsilon > 0$ and any function $g(x)$ continuous over $(x_0 - \epsilon, x_0 + \epsilon)$



Graphical representation of delta function.

Let X be a discrete random variable with range $R_x = \{x_1, x_2, \dots, x_k\}$ and $f(x_i)$, $i=1, 2, \dots, k$ be its probability function. Then, its cumulative function can be written as:

$$F(x) = \sum_{i=1}^k f(x_i) u(x - x_i)$$

which is equivalent to

$$\Rightarrow F(x) = \begin{cases} 0, & x < x_1 \\ f(x_1), & x_1 \leq x \leq x_2 \\ \dots\dots \\ \sum_{i=1}^t f(x_i), & x_t \leq x \leq x_{t+1} \\ \dots\dots \\ 1, & x \geq x_k \end{cases}$$

Then, by differentiating it we have

$$f_G(x) = \frac{dF(x)}{dx} = \sum_{i=1}^k f(x_i) \frac{d}{dx} u(x - x_i) = \sum_{i=1}^k f(x_i) \delta(x - x_i)$$

which is called the generalized probability density function (PDF) of discrete random variable. If $f_G(x)$ does not contain dirac delta, X is continuous. If $f_G(x)$ contains both dirac delta and non-dirac delta functions, X is said to be mixed random variable.

$$f_G(x) = \sum_{i=1}^k f(x_i) \delta(x - x_i) + g(x) \longrightarrow \text{Generalized PDF of a mixed random variable}$$

where $g(x) \geq 0$ does not contain dirac delta function and $f_G(x)$ satisfies.

$$1) f_G(x) \geq 0$$

$$2) \int_{-\infty}^{\infty} f_G(x) dx = \underbrace{\sum_{i=1}^k f(x_i) \int_{-\infty}^{\infty} \delta(x - x_i) dx}_{\sum_{i=1}^k f(x_i)} + \int_{-\infty}^{\infty} g(x) dx = 1$$

Accordingly,

$$\mu_x = E(X) = \int_{-\infty}^{\infty} x f_G(x) dx = \sum_{i=1}^k x_i f(x_i) + \int_{-\infty}^{\infty} x g(x) dx$$

and

$$\text{Var}(X) = \int_{-\infty}^{\infty} (x - \mu_x)^2 f_G(x) dx = \sum_{i=1}^k (x_i - \mu_x)^2 f(x_i) + \int_{-\infty}^{\infty} (x - \mu_x)^2 g(x) dx$$

Example 1: Let X be a continuous random variable with the following probability density function (PDF):

$$f_X(x) = \begin{cases} 2x & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Let also

$$Y = g(X) = \begin{cases} X & 0 \leq X \leq \frac{1}{2} \\ \frac{1}{2} & X > \frac{1}{2} \end{cases}$$

Find the cumulative distribution function (CDF) of Y .

Solution:

First we note that $R_X = [0, 1]$. For $x \in [0, 1]$, $0 \leq g(x) \leq \frac{1}{2}$. Thus, $R_Y = [0, \frac{1}{2}]$, and therefore

$$F_Y(y) = 0, \quad \text{for } y < 0,$$

$$F_Y(y) = 1, \quad \text{for } y > \frac{1}{2}.$$

Now note that

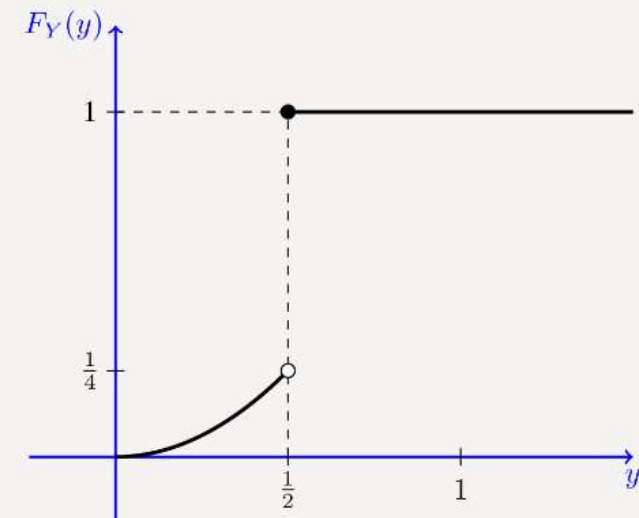
$$\begin{aligned} P\left(Y = \frac{1}{2}\right) &= P\left(X > \frac{1}{2}\right) \\ &= \int_{\frac{1}{2}}^1 2x dx = \frac{3}{4}. \end{aligned}$$

Also, for $0 < y < \frac{1}{2}$,

$$\begin{aligned} F_Y(y) &= P(Y \leq y) \\ &= P(X \leq y) \\ &= \int_0^y 2x dx \\ &= y^2. \end{aligned}$$

Thus, the CDF of Y is given by

$$F_Y(y) = \begin{cases} 1 & y \geq \frac{1}{2} \\ y^2 & 0 \leq y < \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$$



CDF of a Mixed random variable,

The CDF of Y has a continuous part and a discrete part. In particular, we can write

$$F_Y(y) = C(y) + D(y),$$

where $C(y)$ is the continuous part of $F_Y(y)$, i.e.,

$$C(y) = \begin{cases} \frac{1}{4} & y \geq \frac{1}{2} \\ y^2 & 0 \leq y < \frac{1}{2} \\ 0 & y < 0 \end{cases}$$

The discrete part of $F_Y(y)$ is $D(y)$, given by

$$D(y) = \begin{cases} \frac{3}{4} & y \geq \frac{1}{2} \\ 0 & y < \frac{1}{2} \end{cases}$$

Example 2: Let Y be the mixed random variable defined in Example 1.

- a. Find $P(\frac{1}{4} \leq Y \leq \frac{3}{8})$.
- b. Find $P(Y \geq \frac{1}{4})$.
- c. Find EY .

Solution:

Since we have the CDF of Y , we can find the probability that Y is in any given interval. We should pay special attention if the interval includes any jump points.

- a. Find $P(\frac{1}{4} \leq Y \leq \frac{3}{8})$: We can write

$$\begin{aligned} P\left(\frac{1}{4} \leq Y \leq \frac{3}{8}\right) &= F_Y\left(\frac{3}{8}\right) - F_Y\left(\frac{1}{4}\right) + P\left(Y = \frac{1}{4}\right) \\ &= \left(\frac{3}{8}\right)^2 - \left(\frac{1}{4}\right)^2 + 0 = \frac{5}{64}. \end{aligned}$$

- b. Find $P(Y \geq \frac{1}{4})$: We have

$$\begin{aligned} P\left(Y \geq \frac{1}{4}\right) &= 1 - F_Y\left(\frac{1}{4}\right) + P\left(Y = \frac{1}{4}\right) \\ &= 1 - \left(\frac{1}{4}\right)^2 = \frac{15}{16}. \end{aligned}$$

- c. Find EY : Here, we can differentiate the continuous part of the CDF to obtain

$$c(y) = \frac{dC(y)}{dy} = \begin{cases} 2y & 0 \leq y < \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$$

So, we can find EY as

$$\begin{aligned} EY &= \int_0^{\frac{1}{2}} y(2y)dy + \frac{1}{2}P\left(Y = \frac{1}{2}\right) \\ &= \frac{1}{12} + \frac{3}{8} = \frac{11}{24}. \end{aligned}$$



Example: Let X be a random variable with CDF

- a) Plot $F(x)$ and find the type of random variable
- b) Find the PDF of X .
- c) Find $E(e^x)$
- d) Find $P(X = 0 | X \leq 0.5)$

$$F(x) = \begin{cases} 1, & x \geq 1 \\ \frac{1}{2} + \frac{x}{2}, & 0 \leq x < 1 \\ 0, & x < 0 \end{cases}$$



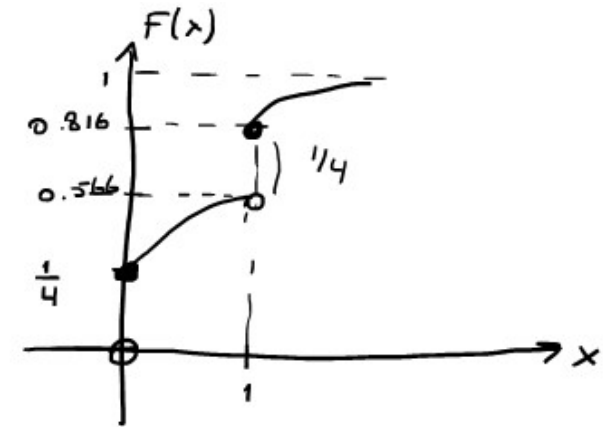
Example: Let X be a random variable with CDF

- a) Plot $F(x)$ and find the type of random variable
- b) Find $f_G(x)$
- c) Find $P(X \leq 3/2)$, $P(X > 1)$ and $P(X \geq 1)$
- d) Compute $E(X)$ and $\text{Var}(X)$.

$$F(x) = \begin{cases} 0, & x < 0 \\ 1/4, & 0 \leq x < 1 \\ \frac{3}{4} + \frac{1}{4}(x-1), & 1 \leq x < 2 \\ 1, & x \geq 2 \end{cases}$$

Example: Let X be a random variable with CDF

$$F(x) = \begin{cases} \frac{1}{2} + \frac{1}{2}(1 - e^{-x}), & x \geq 1 \\ \frac{1}{4} + \frac{1}{2}(1 - e^{-x}), & 0 \leq x < 1 \\ 0, & x < 0 \end{cases}$$



- What kind of variable is X ?
- Find the PDF of X .
- Find $P(X > 0.5)$ both using the CDF and using the PDF.
- Find $E(X)$ and $\text{Var}(X)$.



Example: Let X be a random variable with PDF

$$f(x) = \begin{cases} \frac{1}{4}, & -2 \leq x < 2 \\ 0, & \text{otherwise} \end{cases}$$

We define $Y=g(X)$, where the function $g(x)$ is defined as

$$Y = g(x) = \begin{cases} 1, & x > 1 \\ x, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

$\searrow x < 0$

Find CDF and PDF of Y .

* Note that $R_Y = [0, 1]$

$$F_Y(y) = 0, \quad \text{for } y < 0$$

$$F_Y(y) = 1, \quad \text{for } y > 1$$

$$P(Y=0) = P(X < 0) = \int_{-2}^0 \frac{1}{4} dx = 1/2$$

$$P(Y=1) = P(X > 1) = \int_1^2 \frac{1}{4} dx = 1/4$$

$$0 < y < 1 \Rightarrow F_Y(y) = P(Y \leq y) = P(X \leq y) = F_X(y)$$

The CDF of Y is given by

$$\int_{-2}^y \frac{1}{4} dx = \frac{1}{4}(y+2)$$

$$F_Y(y) = \begin{cases} 1 & , y \geq 1 \\ \frac{y+2}{4} & , 0 \leq y < 1 \\ 0 & , \text{otherwise} \end{cases} \quad \left. \begin{matrix} y=0 \\ y=1 \end{matrix} \right) \text{ Jump points.}$$

$$f_Y(y) = \frac{1}{2} \delta(y) + \frac{1}{4} \delta(y-1) + g(y) \quad \downarrow \quad \frac{1}{4}(u(y) - u(y-1))$$

MULTIVARIATE DISTRIBUTIONS

Two caplets are selected at random from a bottle containing 3 aspirin, 2 sedative, and 4 laxative caplets. If X and Y are, respectively, the numbers of aspirin and sedative caplets included among the 2 caplets drawn from the bottle, find the probabilities associated with all possible pairs of values of X and Y .

JOINT PROBABILITY DISTRIBUTION:

DEFINITION **JOINT PROBABILITY DISTRIBUTION.** *If X and Y are discrete random variables, the function given by $f(x, y) = P(X = x, Y = y)$ for each pair of values (x, y) within the range of X and Y is called the **joint probability distribution** of X and Y .*

THEOREM A bivariate function can serve as the joint probability distribution of a pair of discrete random variables X and Y if and only if its values, $f(x, y)$, satisfy the conditions

1. $f(x, y) \geq 0$ for each pair of values (x, y) within its domain;
2. $\sum_x \sum_y f(x, y) = 1$, where the double summation extends over all possible pairs (x, y) within its domain.

Example: Determine the value of k for which the function given by

$$f(x, y) = kxy \quad \text{for } x = 1, 2, 3; \quad y = 1, 2, 3$$

can serve as a joint probability distribution.

Solution

Substituting the various values of x and y , we get $f(1, 1) = k$, $f(1, 2) = 2k$, $f(1, 3) = 3k$, $f(2, 1) = 2k$, $f(2, 2) = 4k$, $f(2, 3) = 6k$, $f(3, 1) = 3k$, $f(3, 2) = 6k$, and $f(3, 3) = 9k$. To satisfy the first condition of Theorem the constant k must be nonnegative, and to satisfy the second condition,

$$k + 2k + 3k + 2k + 4k + 6k + 3k + 6k + 9k = 1$$

so that $36k = 1$ and $k = \frac{1}{36}$.

JOINT PROBABILITY DENSITY FUNCTION:

DEFINITION **JOINT PROBABILITY DENSITY FUNCTION.** A bivariate function with values $f(x, y)$ defined over the xy -plane is called a *joint probability density function* of the continuous random variables X and Y if and only if

$$P(X, Y) \in A = \iint_A f(x, y) dx dy$$

for any region A in the xy -plane.

THEOREM A bivariate function can serve as a joint probability density function of a pair of continuous random variables X and Y if its values, $f(x, y)$, satisfy the conditions

1. $f(x, y) \geq 0$ for $-\infty < x < \infty$, $-\infty < y < \infty$;
2. $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$.

EXAMPLE

Given the joint probability density function

$$f(x, y) = \begin{cases} \frac{3}{5}x(y+x) & \text{for } 0 < x < 1, 0 < y < 2 \\ 0 & \text{elsewhere} \end{cases}$$

of two random variables X and Y , find $P[(X, Y) \in A]$, where A is the region $\{(x, y) | 0 < x < \frac{1}{2}, 1 < y < 2\}$.

Solution

$$\begin{aligned} P[(X, Y) \in A] &= P\left(0 < X < \frac{1}{2}, 1 < Y < 2\right) \\ &= \int_1^2 \int_0^{\frac{1}{2}} \frac{3}{5}x(y+x) dx dy \\ &= \int_1^2 \left. \frac{3x^2y}{10} + \frac{3x^3}{15} \right|_{x=0}^{x=\frac{1}{2}} dy \\ &= \int_1^2 \left(\frac{3y}{40} + \frac{1}{40} \right) dy = \left. \frac{3y^2}{80} + \frac{y}{40} \right|_1^2 \\ &= \frac{11}{80} \end{aligned}$$

JOINT DISTRIBUTION FUNCTION (JOINT CUMULATIVE DISTRIBUTION):

DEFINITION **JOINT DISTRIBUTION FUNCTION.** If X and Y are discrete random variables, the function given by

$$F(x, y) = P(X \leq x, Y \leq y) = \sum_{s \leq x} \sum_{t \leq y} f(s, t) \quad \text{for } \begin{matrix} -\infty < x < \infty \\ -\infty < y < \infty \end{matrix}$$

where $f(s, t)$ is the value of the joint probability distribution of X and Y at (s, t) , is called the *joint distribution function*, or the *joint cumulative distribution* of X and Y .

Example: For the caplets example, find $F(1, 1)$.

Solution

$$\begin{aligned} F(1, 1) &= P(X \leq 1, Y \leq 1) \\ &= f(0, 0) + f(0, 1) + f(1, 0) + f(1, 1) \\ &= \frac{1}{6} + \frac{2}{9} + \frac{1}{3} + \frac{1}{6} \\ &= \frac{8}{9} \end{aligned}$$

JOINT DISTRIBUTION FUNCTION: (JOINT CUMULATIVE DISTRIBUTION):

DEFINITION JOINT DISTRIBUTION FUNCTION. If X and Y are **continuous** random variables, the function given by

$$F(x, y) = P(X \leq x, Y \leq y) = \int_{-\infty}^y \int_{-\infty}^x f(s, t) ds dt \quad \text{for } -\infty < x < \infty, \\ -\infty < y < \infty$$

where $f(s, t)$ is the joint probability density of X and Y at (s, t) , is called the **joint distribution function** of X and Y .

EXAMPLE

If the joint probability density of X and Y is given by

$$f(x, y) = \begin{cases} x + y & \text{for } 0 < x < 1, 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

find the joint distribution function of these two random variables.

Solution

If either $x < 0$ or $y < 0$, it follows immediately that $F(x, y) = 0$. For $0 < x < 1$ and $0 < y < 1$ (Region I of Figure 10), we get

$$F(x, y) = \int_0^y \int_0^x (s + t) \, ds \, dt = \frac{1}{2}xy(x + y)$$

for $x > 1$ and $0 < y < 1$ (Region II of Figure 10), we get

$$F(x, y) = \int_0^y \int_0^1 (s + t) \, ds \, dt = \frac{1}{2}y(y + 1)$$

for $0 < x < 1$ and $y > 1$ (Region III of Figure 10), we get

$$F(x, y) = \int_0^1 \int_0^x (s + t) \, ds \, dt = \frac{1}{2}x(x + 1)$$

and for $x > 1$ and $y > 1$ (Region IV of Figure 10), we get

$$F(x, y) = \int_0^1 \int_0^1 (s + t) \, ds \, dt = 1$$

Since the joint distribution function is everywhere continuous, the boundaries between any two of these regions can be included in either one, and we can write

$$F(x, y) = \begin{cases} 0 & \text{for } x \leq 0 \text{ or } y \leq 0 \\ \frac{1}{2}xy(x + y) & \text{for } 0 < x < 1, 0 < y < 1 \\ \frac{1}{2}y(y + 1) & \text{for } x \geq 1, 0 < y < 1 \\ \frac{1}{2}x(x + 1) & \text{for } 0 < x < 1, y \geq 1 \\ 1 & \text{for } x \geq 1, y \geq 1 \end{cases}$$

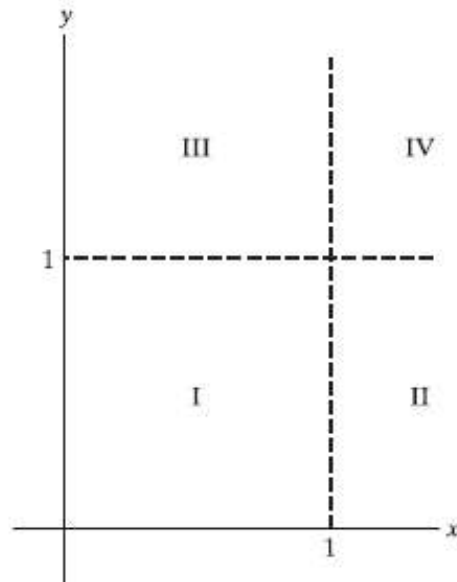
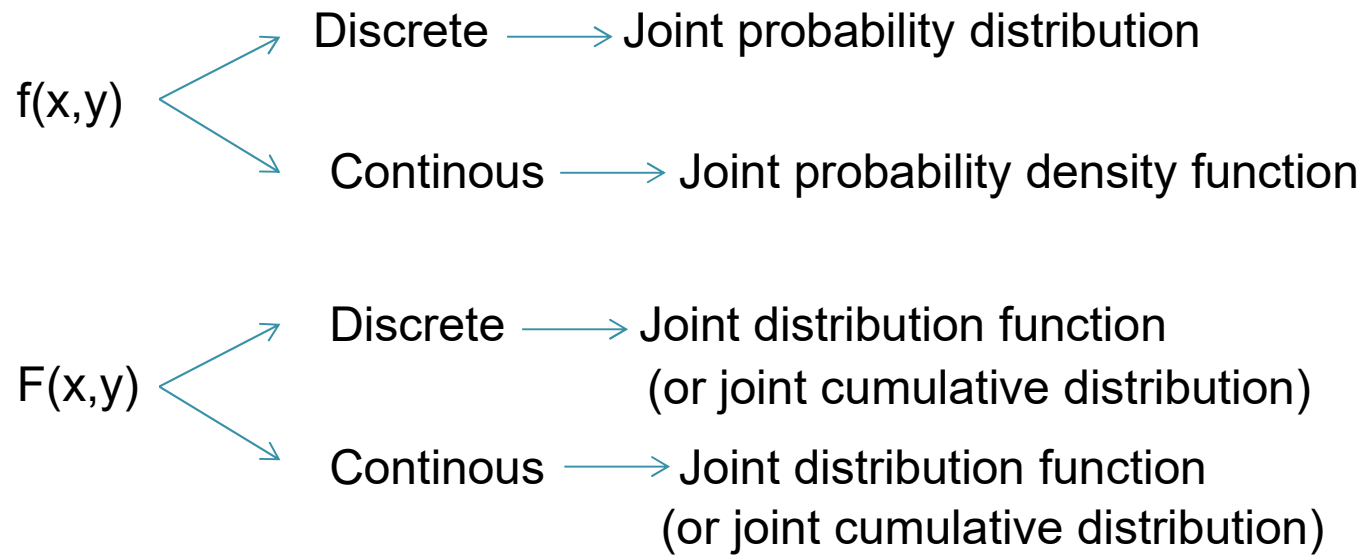


Figure 10. Diagram for Example



$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$$

$$\frac{\partial}{\partial x} F(x, y) = \int_{-\infty}^y f(x, t) \partial t$$

$$\frac{\partial}{\partial y} F(x, y) = \int_{-\infty}^x f(s, y) \partial s$$

EXAMPLE

Find the joint probability density of the two random variables X and Y whose joint distribution function is given by

$$F(x, y) = \begin{cases} (1 - e^{-x})(1 - e^{-y}) & \text{for } x > 0 \text{ and } y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Also use the joint probability density to determine $P(1 < X < 3, 1 < Y < 2)$.

Solution

Since partial differentiation yields

$$\frac{\partial^2}{\partial x \partial y} F(x, y) = e^{-(x+y)}$$

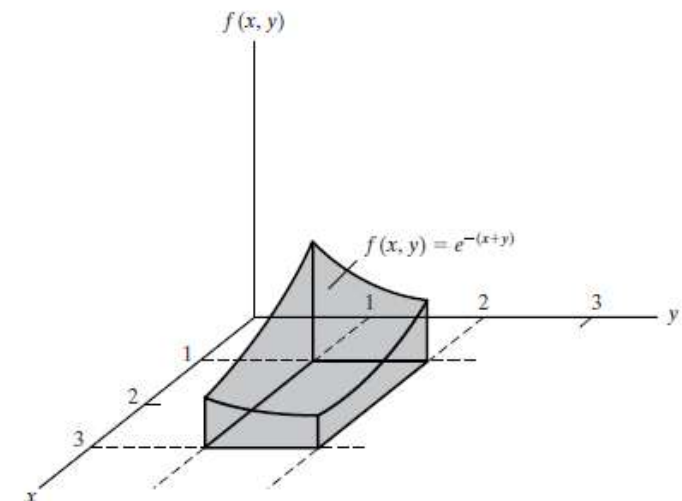
for $x > 0$ and $y > 0$ and 0 elsewhere, we find that the joint probability density of X and Y is given by

$$f(x, y) = \begin{cases} e^{-(x+y)} & \text{for } x > 0 \text{ and } y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Thus, integration yields

$$\begin{aligned} \int_1^2 \int_1^3 e^{-(x+y)} dx dy &= (e^{-1} - e^{-3})(e^{-1} - e^{-2}) \\ &= e^{-2} - e^{-3} - e^{-4} + e^{-5} \\ &= 0.074 \end{aligned}$$

for $P(1 < X < 3, 1 < Y < 2)$.



GENERALIZATION

If $X_1, X_2, \dots, X_n$ are discrete random variables, their joint probability distribution is

$$f(X_1, X_2, \dots, X_n) = P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$$

and their joint distribution function (joint cumulative dist.) is

$$F(X_1, X_2, \dots, X_n) = P(X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n)$$

For the continuous case;

$$F(X_1, X_2, \dots, X_n) = \int_{-\infty}^{x_n} \dots \int_{-\infty}^{x_2} \int_{-\infty}^{x_1} f(t_1, t_2, \dots, t_n) dt_1 dt_2 \dots dt_n$$

and,

$$f(x_1, x_2, \dots, x_n) = \frac{\partial^n}{\partial x_1 \partial x_2 \dots \partial x_n} F(x_1, x_2, \dots, x_n)$$

EXAMPLE

If the joint probability distribution of three discrete random variables X , Y , and Z is given by

$$f(x, y, z) = \frac{(x + y)z}{63} \quad \text{for } x = 1, 2; \quad y = 1, 2, 3; \quad z = 1, 2$$

find $P(X = 2, Y + Z \leq 3)$.

EXAMPLE

If the trivariate probability density of X_1, X_2 , and X_3 is given by

$$f(x_1, x_2, x_3) = \begin{cases} (x_1 + x_2)e^{-x_3} & \text{for } 0 < x_1 < 1, 0 < x_2 < 1, x_3 > 0 \\ 0 & \text{elsewhere} \end{cases}$$

find $P[(X_1, X_2, X_3) \in A]$, where A is the region

$$\left\{ (x_1, x_2, x_3) \mid 0 < x_1 < \frac{1}{2}, \frac{1}{2} < x_2 < 1, x_3 < 1 \right\}$$

MARGINAL DISTRIBUTION:

EXAMPLE

In the caplet example, we derived the joint probability distribution of two random variables X and Y , the number of aspirin caplets and the number of sedative caplets included among two caplets drawn at random from a bottle containing three aspirin, two sedative, and four laxative caplets. Find the probability distribution of X alone and that of Y alone.

Solution

The results of caplet example are shown in the following table, together with the marginal totals, that is, the totals of the respective rows and columns:

		x			
		0	1	2	
y	0	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{12}$	$\frac{7}{12}$
	1	$\frac{2}{9}$	$\frac{1}{6}$		$\frac{7}{18}$
	2	$\frac{1}{36}$			$\frac{1}{36}$
		$\frac{5}{12}$	$\frac{1}{2}$	$\frac{1}{12}$	

The column totals are the probabilities that X will take on the values 0, 1, and 2. In other words, they are the values

$$g(x) = \sum_{y=0}^2 f(x, y) \quad \text{for } x = 0, 1, 2$$

of the probability distribution of X . By the same token, the row totals are the values

$$h(y) = \sum_{x=0}^2 f(x, y) \quad \text{for } y = 0, 1, 2$$

of the probability distribution of Y .

DEFINITION MARGINAL DISTRIBUTION. *If X and Y are discrete random variables and $f(x, y)$ is the value of their joint probability distribution at (x, y) , the function given by*

$$g(x) = \sum_y f(x, y)$$

for each x within the range of X is called the marginal distribution of X . Correspondingly, the function given by

$$h(y) = \sum_x f(x, y)$$

for each y within the range of Y is called the marginal distribution of Y .

When X and Y are continuous random variables, the probability distributions are replaced by probability densities, the summations are replaced by integrals, and we obtain the following definition.

DEFINITION **MARGINAL DENSITY.** *If X and Y are continuous random variables and $f(x, y)$ is the value of their joint probability density at (x, y) , the function given by*

$$g(x) = \int_{-\infty}^{\infty} f(x, y) dy \quad \text{for } -\infty < x < \infty$$

is called the marginal density of X . Correspondingly, the function given by

$$h(y) = \int_{-\infty}^{\infty} f(x, y) dx \quad \text{for } -\infty < y < \infty$$

is called the marginal density of Y .

EXAMPLE

Given the joint probability density

$$f(x, y) = \begin{cases} \frac{2}{3}(x + 2y) & \text{for } 0 < x < 1, 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

find the marginal densities of X and Y .

When we are dealing with more than two random variables, we can speak not only of the marginal distributions of the individual random variables, but also of the **joint marginal distributions** of several of the random variables. If the joint probability distribution of the discrete random variables $X_1, X_2, \dots$, and X_n has the values $f(x_1, x_2, \dots, x_n)$, the marginal distribution of X_1 alone is given by

$$g(x_1) = \sum_{x_2} \cdots \sum_{x_n} f(x_1, x_2, \dots, x_n)$$

for all values within the range of X_1 , the joint marginal distribution of X_1, X_2 , and X_3 is given by

$$m(x_1, x_2, x_3) = \sum_{x_4} \cdots \sum_{x_n} f(x_1, x_2, \dots, x_n)$$

for all values within the range of X_1, X_2 , and X_3 , and other marginal distributions can be defined in the same way. For the continuous case, probability distributions are replaced by probability densities, summations are replaced by integrals, and if the joint probability density of the continuous random variables $X_1, X_2, \dots$, and X_n has the values $f(x_1, x_2, \dots, x_n)$, the marginal density of X_2 alone is given by

$$h(x_2) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x_1, x_2, \dots, x_n) dx_1 dx_3 \cdots dx_n$$

for $-\infty < x_2 < \infty$, the joint marginal density of X_1 and X_n is given by

$$\varphi(x_1, x_n) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x_1, x_2, \dots, x_n) dx_2 dx_3 \cdots dx_{n-1}$$

for $-\infty < x_1 < \infty$ and $-\infty < x_n < \infty$, and so forth.

EXAMPLE

Considering again the trivariate probability density of Example 1

$$f(x_1, x_2, x_3) = \begin{cases} (x_1 + x_2)e^{-x_3} & \text{for } 0 < x_1 < 1, 0 < x_2 < 1, x_3 > 0 \\ 0 & \text{elsewhere} \end{cases}$$

find the joint marginal density of X_1 and X_3 and the marginal density of X_1 alone.

Solution

Performing the necessary integration, we find that the joint marginal density of X_1 and X_3 is given by

$$m(x_1, x_3) = \int_0^1 (x_1 + x_2)e^{-x_3} dx_2 = \left(x_1 + \frac{1}{2}\right)e^{-x_3}$$

for $0 < x_1 < 1$ and $x_3 > 0$ and $m(x_1, x_3) = 0$ elsewhere. Using this result, we find that the marginal density of X_1 alone is given by

$$\begin{aligned} g(x_1) &= \int_0^\infty \int_0^1 f(x_1, x_2, x_3) dx_2 dx_3 = \int_0^\infty m(x_1, x_3) dx_3 \\ &= \int_0^\infty \left(x_1 + \frac{1}{2}\right)e^{-x_3} dx_3 = x_1 + \frac{1}{2} \end{aligned}$$

for $0 < x_1 < 1$ and $g(x_1) = 0$ elsewhere.